

Practice Test I (Version 1)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) Calculators are not allowed for this test.
- V) Each problem is worth four points for a total of 100 points.

Determine if the expressions below are polynomials. If not, circle the term(s) that disqualify each expression.

① i) $8x^4 - 2$

Yes

ii) $6 - x + 3x^2 + 7x - 1$

No

iii) $3x^{-4} + 2\sqrt{x} - 1$

No

iv) $x^5 - x^4 + x^3 - x^2 - x + 10$

Yes

② Add the polynomials below.

$$(x^3 - 5x^2 + 2) + (-3x^4 + 2x^3 - 8x^2)$$

$$\boxed{-3x^4 + 3x^3 - 13x^2 + 2}$$

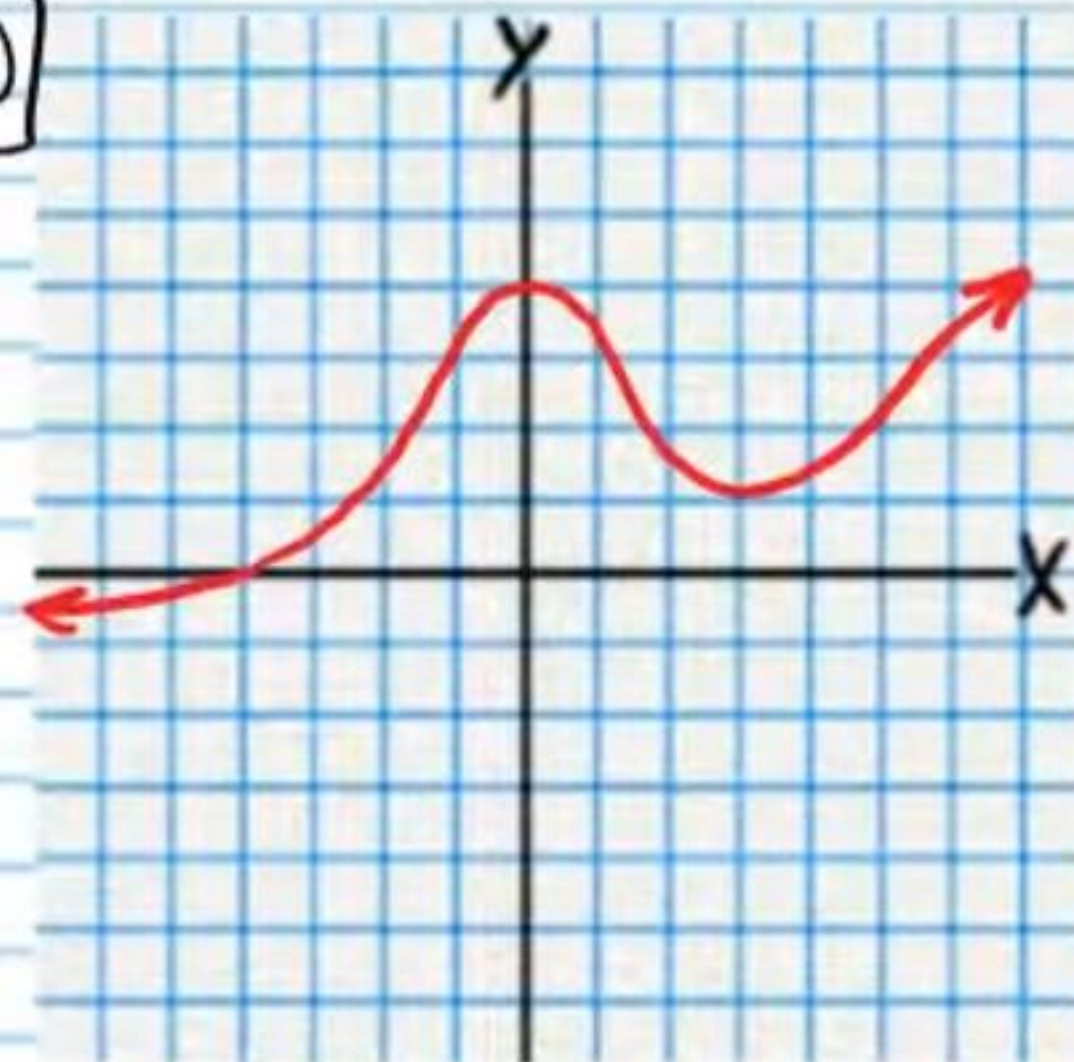
③ Divide.

$$\frac{27x^4 - 81x^3 + 30x^2}{3x^2} = \boxed{9x^2 - 27x + 10}$$

④ The graph of a function is written below. Write the intervals where the function is increasing or decreasing.

Increasing: $(-7, 0)$ and $(3, 7)$

Decreasing: $(0, 3)$



⑤ Use the Rational Zeros Theorem to find all of the potential rational zeros of the function below.

$$f(x) = 3x^5 + 9x^4 + 2x^2 - 4x + 2$$

$$p: 2 \Rightarrow 1, 2$$

$$q: 3 \Rightarrow 1, 3$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm \frac{1}{3}, \pm \frac{2}{3}$$

⑥ What is the maximum number of turning points of the graph of the function in problem #5?

4

⑦ Indicate the end behavior of the polynomial in problem #5.

$$\text{As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

⑧ Write the zeros of the polynomial below, the multiplicity of each zero, and whether or not the corresponding graph crosses or simply touches the x-axis at each zero.

$$f(x) = 2(x-1)^6(x+2)^3$$

$x = 1 \Rightarrow$ Multiplicity: 6 Touches the x-axis.

$x = -2 \Rightarrow$ Multiplicity: 3 Crosses the x-axis

⑨ Divide using synthetic division.

$$(3x^5 + 2x^2 + 4x^4) \div (-2 + x)$$

$$(3x^5 + 4x^4 + 2x^2) \div (x - 2)$$

$$\begin{array}{r|rrrrrr} 2 & 3 & 4 & 0 & 2 & 0 & 0 \\ & & 6 & 20 & 40 & 84 & 168 \\ \hline & 3 & 10 & 20 & 42 & 84 & 168 \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ & 3x^4 & +10x^3 & +20x^2 & +42x & +84 & +\frac{168}{x-2} \end{array}$$

⑩ Find the zeros of the following function.

$$f(x) = 4x^4 - 29x^2 + 25$$

$$= (4x^2 - 25)(x^2 - 1)$$

$$= (2x - 5)(2x + 5)(x + 1)(x - 1)$$

$$x = \frac{5}{2} \text{ or } x = -\frac{5}{2} \text{ or } x = -1 \text{ or } x = 1$$

⑪ Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

$$2 + 3i, 2 \quad LC = -2, D = 3$$

$$f(x) = -2(x - 2)(x - (2 + 3i))(x - (2 - 3i))$$

$$f(x) = -2(x - 2)(x - 2 - 3i)(x - 2 + 3i)$$

$$f(x) = -2(x - 2)(x^2 - 2x + 3xi - 2x + 4 - 6i - 3xi + 6i + 9)$$

$$f(x) = -2(x - 2)(x^2 - 4x + 13)$$

$$f(x) = -2(x^3 - 4x^2 + 13x - 2x^2 + 8x - 26)$$

$$f(x) = -2(x^3 - 6x^2 + 21x - 26)$$

$$f(x) = -2x^3 + 12x^2 - 42x + 52$$

Solve. Write the solutions graphically and in interval notation.

$$(12) \frac{x}{x^3-64} \leq \frac{3}{x^3-64} + \frac{2x}{x^3-64}$$

$$\frac{x}{x^3-64} - \frac{x}{x^3-64} \leq \frac{3}{x^3-64} + \frac{2x}{x^3-64} - \frac{x}{x^3-64}$$

$$0 \leq \frac{3}{x^3-64} + \frac{x}{x^3-64}$$

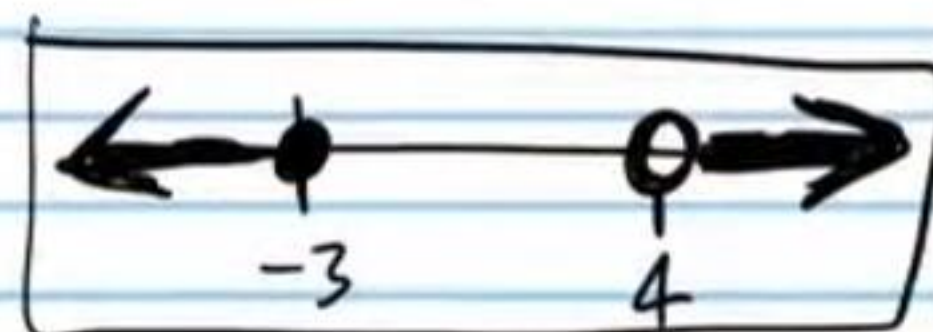
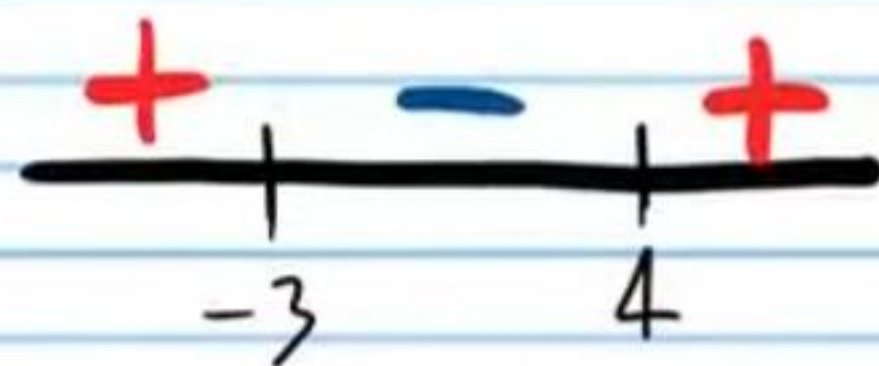
$$0 \leq \frac{x+3}{x^3-64}$$

$$0 \leq \frac{x+3}{x^3-4^3}$$

$$0 \leq \frac{x+3}{(x-4)(x^2+4x+16)}$$

$$0 \leq \frac{x+3}{(x-4)(x^2+4x+16)}$$

$$\begin{aligned} D &= b^2 - 4ac = 4^2 - 4(1)(16) \\ &= 16 - 64 \\ &= -48 \end{aligned}$$



$$(-\infty, -3] \cup (4, \infty)$$

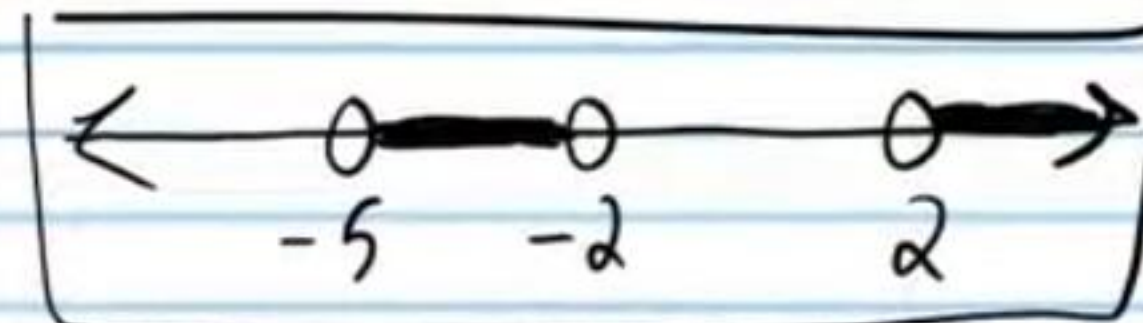
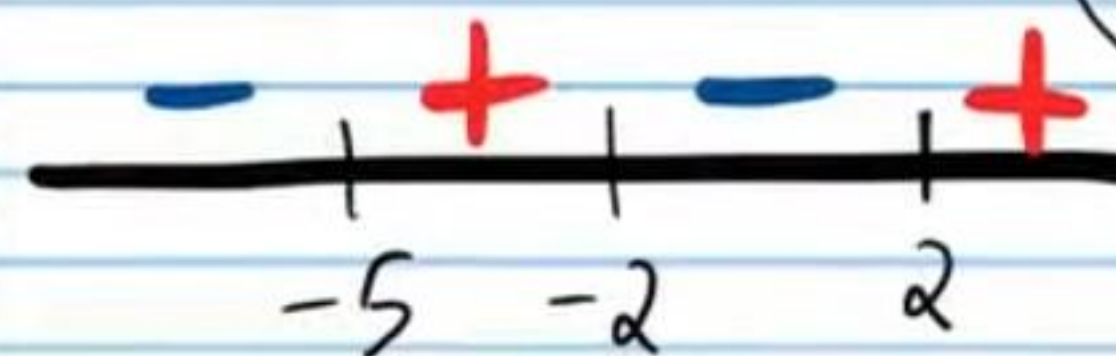
$$(13) x^3 + 5x^2 > 4x + 20$$

$$x^3 + 5x^2 - 4x - 20 > 0$$

$$x^2(x+5) - 4(x+5) > 0$$

$$(x+5)(x^2-4) > 0$$

$$(x+5)(x+2)(x-2) > 0$$

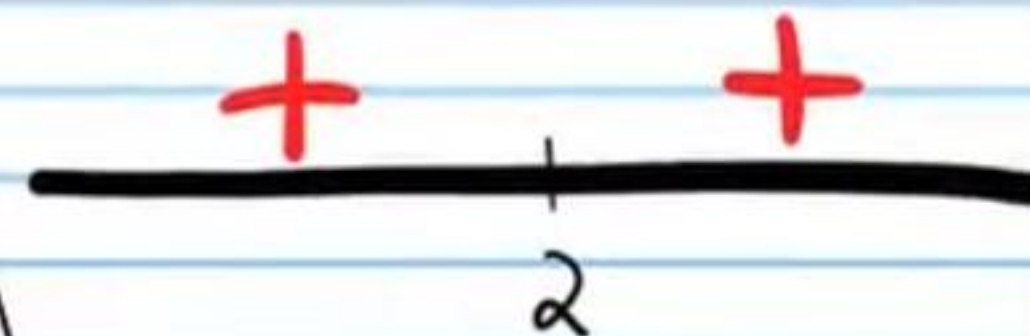


$$(-5, -2) \cup (2, \infty)$$

$$(14) x^2 + 4 \leq 4x$$

$$x^2 - 4x + 4 \leq 0$$

$$(x-2)(x-2) \leq 0$$



$$[2, 2]$$

$$(15) \quad 2 < \frac{x+1}{x+3}$$

$$0 < \frac{x+1}{x+3} - 2$$

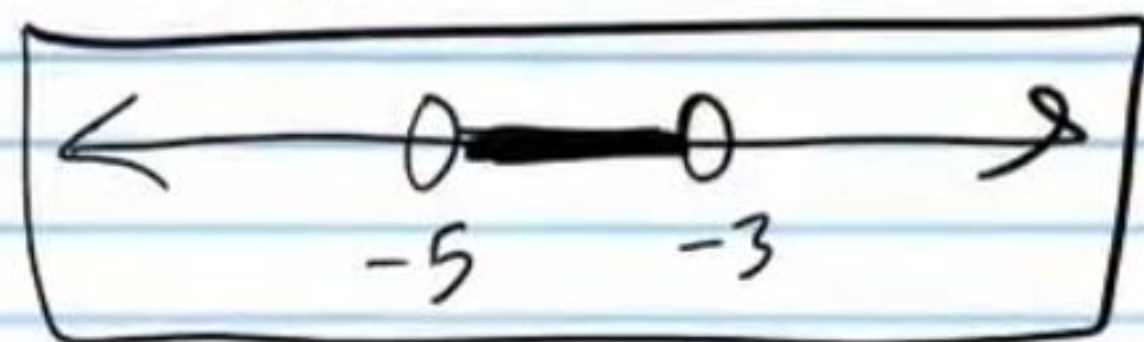
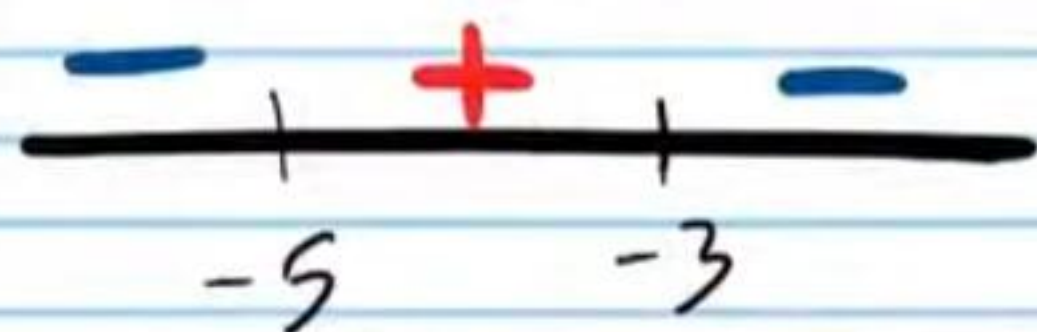
$$0 < \frac{x+1}{x+3} - \frac{2}{1}$$

$$0 < \frac{x+1}{x+3} - \frac{2(x+3)}{1(x+3)}$$

$$0 < \frac{x+1}{x+3} - \frac{2x+6}{x+3}$$

$$0 < \frac{-x-5}{x+3}$$

$$0 < -\frac{x+5}{x+3}$$



$$(-5, -3)$$

Graph the following functions. Draw asymptotes as dotted lines and write their equations.

$$(16) \quad f(x) = \frac{x^2 - 4x + 3}{x^2 + 2x - 8} \quad \text{Horizontal Asymptote Equation}$$

$$= \frac{(x-3)(x-1)}{(x+4)(x-2)}$$

$$x^2 + 2x - 8 \overline{) x^2 - 4x + 3}$$

$$\underline{-6x + 11}$$

$$f(x) = 1 + \frac{-6x + 11}{x^2 + 2x - 8}$$

$$y = 1$$

Vertical Asymptote Equation(s)

$$x = -4, x = 2$$

Y-Intercept

$$f(0) = -\frac{3}{8}$$

$$(0, -\frac{3}{8})$$

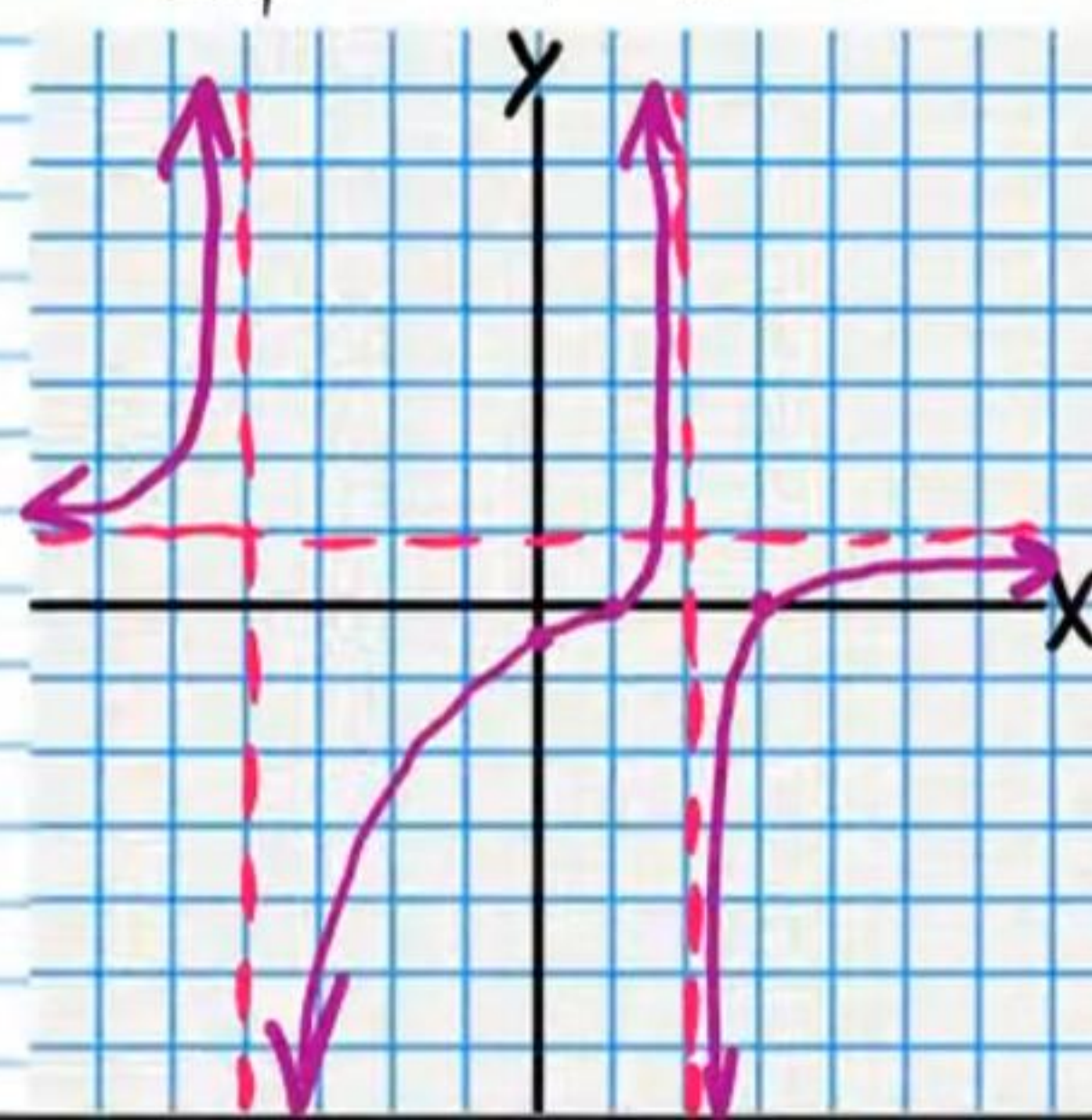
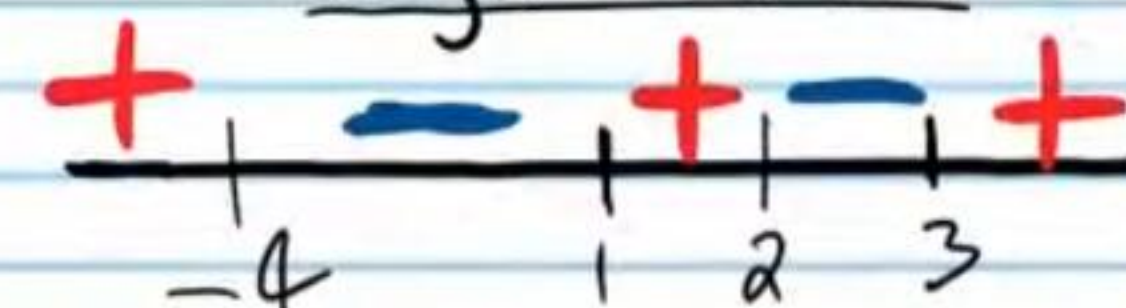
X-Intercept

$$f(x) = 0$$

$$\frac{(x-3)(x-1)}{(x+4)(x-2)} = 0$$

$$(3, 0), (1, 0)$$

Sign Chart



$$\textcircled{17} f(x) = \frac{x-3}{x^2-9}$$

Hole

$$f(x) = \frac{x-3}{(x-3)(x+3)}$$

$$x=3$$

$$g(x) = \frac{1}{x+3}$$

Vertical Asymptote Equations

$$x=-3$$

Horizontal Asymptote Equations

$$y=0$$

Y-Intercept

$$f(0) = \frac{1}{3}$$

$$(0, \frac{1}{3})$$

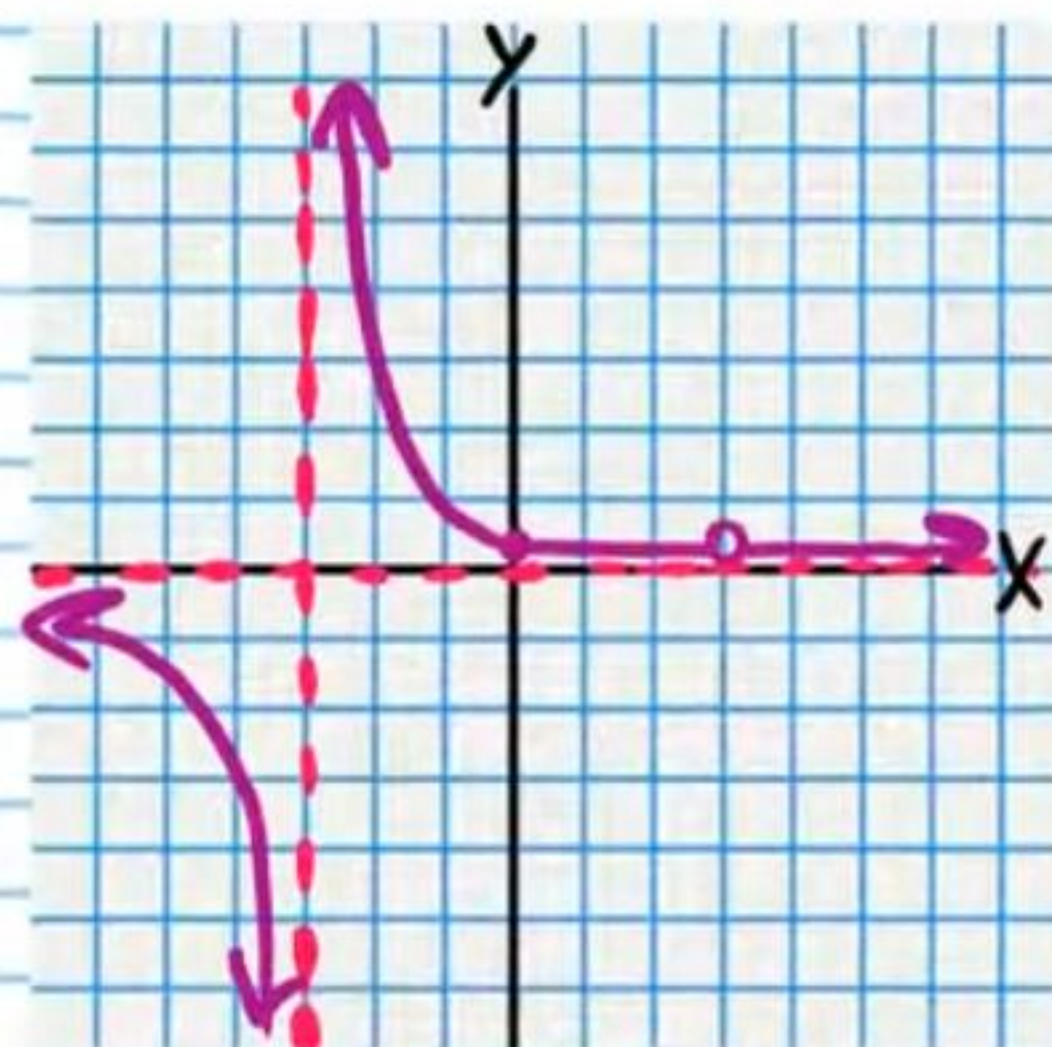
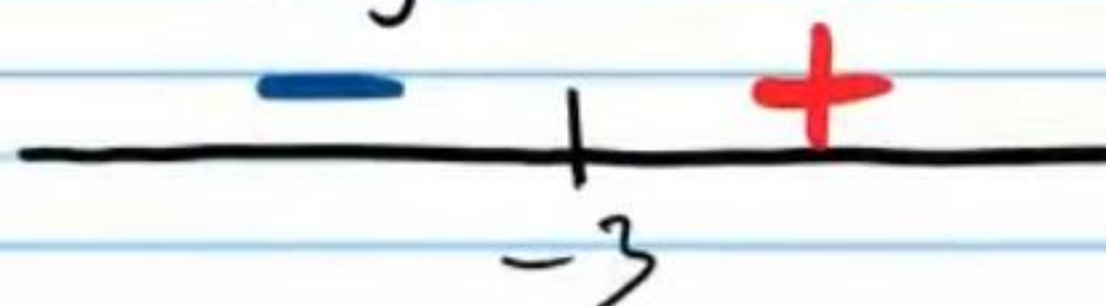
X-Intercept

$$f(x) = 0$$

$$\frac{x-3}{x^2-9} = 0$$

$$\frac{1}{x+3} = 0 \quad \text{None}$$

Sign Chart



$$\textcircled{18} f(x) = \frac{x+1}{x^2-x-12}$$

Vertical Asymptote Equations

$$x=4, x=-3$$

$$f(x) = \frac{x+1}{(x-4)(x+3)}$$

Horizontal Asymptote Equations

$$y=0$$

Y-Intercept

$$f(0) = -\frac{1}{12}$$

$$(0, -\frac{1}{12})$$

X-Intercept

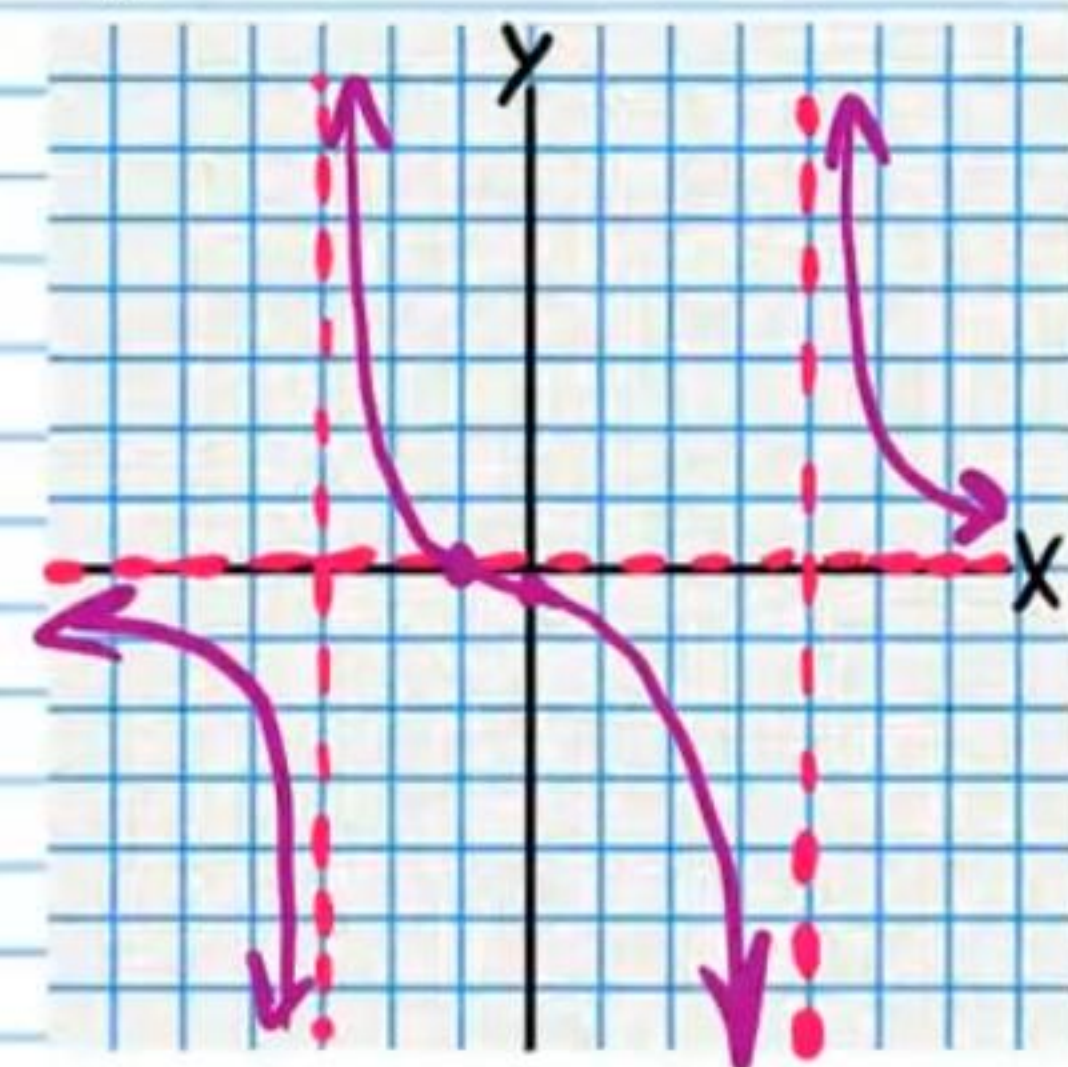
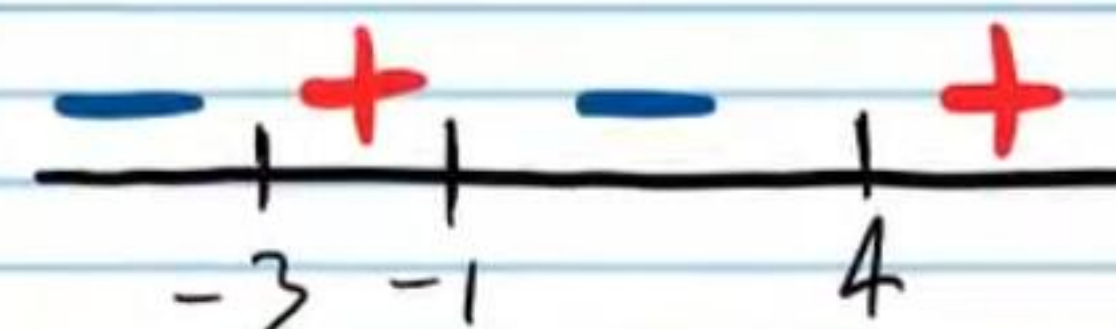
$$f(x) = 0$$

$$\frac{x+1}{(x-4)(x+3)} = 0$$

$$x=-1$$

$$(-1, 0)$$

Sign Chart



①9 Divide.

$$(2x^6) \div (x^2 + x - 2)$$

$$\begin{array}{r}
 x^2 + x - 2 \overline{) 2x^6} \\
 \underline{2x^6 + 2x^5 - 4x^4} \\
 -2x^5 + 4x^4 \\
 \underline{-2x^5 - 2x^4 + 4x^3} \\
 6x^4 - 4x^3 \\
 \underline{6x^4 + 6x^3 - 12x^2} \\
 -10x^3 + 12x^2 \\
 \underline{-10x^3 - 10x^2 + 20x} \\
 22x^2 - 20x \\
 \underline{22x^2 + 22x - 44} \\
 -42x + 44
 \end{array}$$

$$2x^4 - 2x^3 + 6x^2 - 10x + 22 + \frac{-42x + 44}{x^2 + x - 2}$$

②0 Find the roots of the polynomial function below and write the function in factored form.

$$f(x) = x^4 - 3x^3 - 15x^2 + 19x + 30$$

$$P: 30 \Rightarrow 1, 2, 3, 5, 6, 10, 15, 30$$

$$q: 1 \Rightarrow 1$$

$$\frac{P}{q}: \pm 1, \pm 2, \pm 3, \pm 5 \dots$$

$$x = 1 \Rightarrow \text{X} \quad x = -1 \Rightarrow \checkmark$$

$$x = -1 \Rightarrow \text{X} \quad x = 2 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrrr}
 -1 & 1 & -3 & -15 & 19 & 30 \\
 & & -1 & 4 & 11 & -30 \\
 \hline
 & 1 & -4 & -11 & 30 & 0 \\
 & \downarrow & \downarrow & \downarrow & \downarrow & \\
 & x^3 & -4x^2 & -11x & +30 &
 \end{array}$$

$$\begin{array}{r|rrrr}
 2 & 1 & -4 & -11 & 30 \\
 & & 2 & -4 & -30 \\
 \hline
 & 1 & -2 & -15 & 0
 \end{array}$$

$$x^2 - 2x - 15$$

$$f(x) = (x+1)(x-2)(x^2 - 2x - 15)$$

$$f(x) = (x+1)(x-2)(x-5)(x+3)$$

$$-1, 2, 5, -3$$

21) Divide.

$$(x^5 - 4x^4 - 9x^3 + 32x^2 + 28x - 48) \div (x^2 + 4x + 4)$$

$$\begin{array}{r} x^3 - 8x^2 + 19x - 12 \\ x^2 + 4x + 4 \overline{) x^5 - 4x^4 - 9x^3 + 32x^2 + 28x - 48} \\ \underline{x^5 + 4x^4 + 4x^3} \\ -8x^4 - 13x^3 + 32x^2 + 28x - 48 \\ \underline{-8x^4 - 32x^3 - 32x^2} \\ 19x^3 + 64x^2 + 28x - 48 \\ \underline{19x^3 + 76x^2 + 76x} \\ -12x^2 - 48x - 48 \\ \underline{-12x^2 - 48x - 48} \\ 0 \end{array}$$

$$\boxed{x^3 - 8x^2 + 19x - 12}$$

22) Find the zeros of the function below. Write all of them, whether rational, irrational, or imaginary.

$$f(x) = x^6 - 2x^5 + 4x^4 - 6x^3 - 15x^2 + 36x - 18$$

$$p: 18 \Rightarrow 1, 2, 3, 6, 9, 18$$

$$q: 1 \Rightarrow 1$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18$$

$$x = 1 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrrrrr} 1 & 1 & -2 & 4 & -6 & -15 & 36 & -18 \\ & & 1 & -1 & 3 & -3 & -18 & 18 \\ \hline & 1 & -1 & 3 & -3 & -18 & 18 & 0 \end{array}$$

$$x = 1 \Rightarrow \checkmark$$

$$x^5 - x^4 + 3x^3 - 3x^2 - 18x + 18$$

$$\begin{array}{r|rrrrrr} 1 & 1 & -1 & 3 & -3 & -18 & 18 \\ & & 1 & 0 & 3 & 0 & -18 \\ \hline & 1 & 0 & 3 & 0 & -18 & 0 \end{array}$$

$$x^4 + 3x^2 - 18$$

$$f(x) = (x-1)(x-1)(x^4 + 3x^2 - 18)$$

$$f(x) = (x-1)^2(x^2 - 3)(x^2 + 6)$$

$$x^2 - 3 = 0 \text{ or } x^2 + 6 = 0$$

$$x = \pm\sqrt{3} \text{ or } x = \pm i\sqrt{6}$$

$$\boxed{1, \sqrt{3}, -\sqrt{3}, i\sqrt{6}, -i\sqrt{6}}$$

23 Find the x-intercepts of the graph of the function below and determine if the graph crosses or touches the x-axis at each intercept.

$$f(x) = 5x^3 + 5x^2 - 80x + 100$$

$$p: 100 \Rightarrow 1, 2, 4, 5, 10, 20, 25, 50, 100$$

$$q: 5 \Rightarrow 1, 5$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm 4, \pm 5 \dots$$

$$x=1 \Rightarrow \text{X} \quad x=-1 \Rightarrow \text{X} \quad x=2 \checkmark$$

$$\begin{array}{r|rrrr} 2 & 5 & 5 & -80 & 100 \\ & & 10 & 30 & -100 \\ \hline & 5 & 15 & -50 & 0 \\ & \downarrow & \downarrow & \downarrow & \\ & 5x^2 & +15x & -50 & \end{array}$$

$$f(x) = (x-2)(5x^2+15x-50)$$

$$= 5(x-2)(x^2+3x-10)$$

$$= 5(x-2)(x+5)(x-2)$$

$$= 5(x-2)^2(x+5)$$

$2 \Rightarrow$ Touches x-axis

$-5 \Rightarrow$ crosses x-axis

24 For the function below, find the information requested. *It is not necessary to draw its graph.*

$$f(x) = \frac{2x^5 - 5x^4}{2x^3 - x^2 - 10x}$$

Hole(s)

$$x = \frac{5}{2}$$

$$f(x) = \frac{x^4(2x-5)}{x(2x^2-x-10)}$$

$$x=0$$

$$f(x) = \frac{x \cdot x \cdot x \cdot x \cdot (2x-5)}{x \cdot (2x-5) \cdot (x+2)}$$

Vertical Asymptote Equation(s)

$$x = -2$$

Asymptote Curve

$$\begin{array}{r} x^2 - 2x + 4 \\ 2x^3 - x^2 - 10x \overline{) 2x^5 - 5x^4} \\ \underline{2x^5 - x^4 - 10x^3} \\ -4x^4 + 10x^3 \\ \underline{-4x^4 + 2x^3 + 20x^2} \\ 8x^3 - 20x^2 \\ \underline{8x^3 - 4x^2 - 40x} \\ -16x^2 + 40x \end{array}$$

$$f(x) = x^2 - 2x + 4 + \frac{-16x^2 + 40x}{2x^3 - x^2 - 10x}$$

$$x^2 - 2x + 4$$

Y-Intercept

None

X-Intercept

None

②5 Find the solutions of the equation below.

$$0 = 9x^4 - 57x^3 + 131x^2 - 129x + 46$$

$$p: 46 \Rightarrow 1, 2, 23, 46$$

$$q: 9 \Rightarrow 1, 3, 9$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm 23, \pm 46, \pm \frac{1}{3}, \pm \frac{2}{3} \dots$$

$$x=1 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrrr} 1 & 9 & -57 & 131 & -129 & 46 \\ & & 9 & -48 & 83 & -46 \\ \hline & 9 & -48 & 83 & -46 & 0 \end{array}$$

$$x=1 \Rightarrow \times$$

$$9x^3 - 48x^2 + 83x - 46$$

$$x=-1 \Rightarrow \times$$

$$x=2 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrr} 2 & 9 & -48 & 83 & -46 \\ & & 18 & -60 & 46 \\ \hline & 9 & -30 & 23 & 0 \end{array}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $9x^2 - 30x + 23$

$$f(x) = (x-1)(x-2)(9x^2 - 30x + 23)$$

$$f(x) = (x-1)(x-2)(9x^2 - 30x + 23)$$

$$\begin{aligned} D &= b^2 - 4ac = (-30)^2 - 4(9)(23) \\ &= 900 - 828 \\ &= 72 \end{aligned}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-30) \pm \sqrt{72}}{2(9)}$$

$$x = \frac{30 \pm 6\sqrt{2}}{18}$$

$$x = \frac{5 \pm \sqrt{2}}{3}$$

$$x=1 \text{ or } x=2 \text{ or } x = \frac{5+\sqrt{2}}{3} \text{ or } x = \frac{5-\sqrt{2}}{3}$$